

Progress not Regress

Progress Bar Types

- **Indeterminate Progress Bar:** Shows ongoing activity along a track when total progress time is unknown, making it ideal for background downloads, server syncs, and file processing.
- **Splash Screen:** Introduces the application while core resources load into memory on startup, best used during app launches and mobile app initialization.
- **Skeleton Screen:** Acts as a placeholder with a simplified layout of a webpage using grey shapes, mimicking the final content to reduce cognitive load when wait times are under ten seconds.
- **Throbber:** A type of indeterminate progress bar (like a spinning wheel) that indicates active processing without displaying percentages, though long waits can frustrate users.

Economic & Social Indicators

- **Gross Domestic Product (GDP):** Measures the total market value of goods produced by a country using consumption, spending, net exports, and investment; it factors in Purchasing Power Parity (PPP) for accurate cross-country comparisons but does not directly reflect standard of living.
- **Gross National Income (GNI):** Calculates total income earned by residents (GDP plus net income from non-residents); GNI is nearly equal to GDP in developed nations, while foreign aid often makes it larger in developing countries.
- **Gini Coefficient:** Developed by Corrado Gini to measure inequality in wealth distribution, where a score of 0 represents perfect equality and 1 represents maximum inequality, with Slovakia holding the lowest globally.
- **Human Development Index (HDI):** Created by Pakistani economist Mahbub ul Haq, this composite index evaluates life expectancy, education, and per capita income, though the Inequality-adjusted HDI (IHDI) provides a more precise measurement.
- **Multidimensional Poverty Index (MPI):** Developed by the UNDP, this flexible method assesses overall poverty through health, education, and monetary indicators, where higher figures signify greater poverty levels.
- **Labor Productivity:** Quantifies overall economic output produced per hour worked.

- **Happiness Index:** Ranks countries in the World Happiness Report based on several quality-of-life factors.
- **Henley Passport Index:** Measures global travel freedom by ranking passports according to the number of destinations accessible visa-free.
- **Big Mac Index:** An informal metric created to compare purchasing power parity between currencies based on the relative cost of a McDonald's Big Mac.

Placebo Buttons & Behavioral Psychology

- **Placebo Controls:** Non-functional UI elements that provide an illusion of control, effectively reducing user stress, encouraging patience, and minimizing complaints.
- **Office Thermostats:** Dummy HVAC units installed in offices—some venting pneumatic air into empty beams—provide a tactile response to comfort concerns, statistically cutting maintenance requests.
- **Pedestrian Crosswalk Buttons:** Many city walk buttons (such as 2,500 out of 3,250 in NYC) are deactivated in favor of automated timers, but keeping the physical button maintains the signal-request ritual to prevent unsafe crossing.
- **Elevator "Close Door" Buttons:** Often deactivated for public use in post-1990 US installations and reserved for emergency personnel keys, allowing passengers the feeling of speeding up automated safety timers.
- **Endowed Progress Effect:** States that people are more motivated to complete a goal when they feel progress has already begun, explaining why progress bars that start fast and slow down near the end are psychologically effective.
- **Houston Airport Baggage Handling:** Resolved passenger complaints about long luggage wait times by making the walk from the arrival gate longer, shifting time spent waiting idle at the carousel into active walking.
- **"Surely the World's Getting Better":** An artwork by Will Crawford illustrating the evolution of warfare from naval battles to modern multi-domain conflicts, expressing the artist's view that humanity does not improve over time.
- **Namco Loading Screen Patent:** Video game company Namco held a famous patent on playable mini-games during initial loading screens, notably utilized in titles like *Ridge Racer*.

More To Do than can ever be listed

Historical and psychological foundations:

The practice of documenting tasks is a centuries-old tradition. It serves as a bridge between abstract goals and actionable steps.

Historical precedent: Renowned thinkers and leaders, including Leonardo da Vinci and Benjamin Franklin, utilized written lists to clarify priorities and ensure the completion of complex projects.

Psychological drivers: Research indicates a specific cognitive pattern where the human brain prioritizes the memory of unfinished tasks over completed ones. Written lists act as an external memory aid, relieving the brain of the burden of constant recall and allowing for deeper focus on the task at hand.

Fulfillment vs Frustration: When lists are excessively long or lack organization, they become sources of stress. However, when used as a tool for clarification, they contribute to a sense of fulfillment and productivity.

Task Management Methods:

Eisenhower method: Categorizes tasks into four boxes based on Urgent vs. Important to ensure focus on meaningful work.

1-2-3 Method: Limits daily goals to one primary point, two secondary priorities, and three tertiary priorities.

Ivy Lee Method: Requires listing exactly six tasks for the next day; must be completed in strict order.

4 D method: Filters tasks into four immediate actions: Do, delegate, defer, or delete.

Eat That Frog: Prioritizes the most challenging task at the start of the day to ease subsequent work.

Must-do list: Strictly limit to non-optional tasks to eliminate distractions and reduce overwhelm.

Bullet Journal: A flexible system combining a diary and planner using specific bullets and migration of tasks.

Zeigarnik effect:

It's easier to remember things we didn't finish than remember the things we did. This is called the Zeigarnik effect.

The end is Nearish

Government & Political Concepts

- **Caretaker Government:** A temporary administration that manages day-to-day governance during election periods without making major policy decisions, a practice restored by Bangladesh in 2025.
- **The Doomsday Clock:** Created by the Bulletin of the Atomic Scientists, this symbolic tracker measures humanity's proximity to global annihilation by factoring in threats such as nuclear war and climate change, setting the clock to 89 seconds to midnight in 2025.
- **Lame Ducks:** Refers to political leaders approaching the end of their second term whose influence wanes; Queen Elizabeth II famously delayed announcing her successor to keep public focus on the present rather than future transitions.

Historical Apocalyptic Predictions & Panics

- **Year 1666 Panic:** Driven by the apocalyptic number of the beast (666) and a millennium since Christ's appearance, many predicted the end of the world in 1666, a year that coincided with the Great Fire of London.
- **Halley's Comet Panic (1910):** The 1910 passage of Halley's Comet sparked widespread fear that its tail would release toxic gases or collide with Earth, causing mass panic across the globe.
- **The Prophet Hen of Leeds (1806):** Mass unrest was created when a hen in Leeds appeared to lay eggs inscribed with "Christ is coming," though it was later revealed to be a hoax where the owner etched the eggs and reinserted them.
- **The Millerites:** Preacher William Miller predicted the Second Coming for 1843 and rescheduled it to 1844, leading to widespread disillusionment known as the Great Disappointment when it failed to occur.
- **Wodziwob's Visions:** A Native American prophet who envisioned a massive earthquake that would consume non-indigenous populations while restoring Native Americans to live freely on their lands.
- **The Jupiter Effect:** A 1974 book by John Gribbin and Stephen Plagemann predicting that a grand planetary alignment on March 10, 1982, would trigger massive natural disasters along the San Andreas Fault, which failed to materialize.
- **Chen Tao (Hon-Ming Chen):** A Taiwanese UFO cult led by Hon-Ming Chen that relocated to Texas, whose members wore white suits and cowboy hats while making

predictions about extraterrestrial salvation.

- **The Y2K Bug:** Fear centered on older computer systems storing years as two digits, causing concerns that the year 2000 would be interpreted as 1900 and crash global infrastructure, though actual disruptions proved negligible.
- **2012 Mayan Apocalypse:** A popular misconception held that the conclusion of the Mayan Long Count calendar on December 21, 2012, signified the end of the world, whereas it traditionally represented the start of a new spiritual cycle.

Apocalyptic & End-Times Works in Art & Media

- ***Four Horsemen of the Apocalypse* (Albrecht Dürer):** A woodcut depicting Death, Famine, War, and Plague trampling humanity, based directly on the Book of Revelation.
- ***Mystic Nativity* (Sandro Botticelli):** Shows the infant Christ and Mary surrounded by angels alongside fleeing demons, featuring a Greek inscription referencing the French invasion of Italy to draw parallels to biblical turmoil.
- ***The Last Judgment* (Michelangelo):** A monumental Sistine Chapel fresco depicting the final sorting of souls at the end of the world, centered on Christ and incorporating imagery from Dante's *Inferno*.
- ***The Triumph of Death* (Pieter Bruegel the Elder):** A panoramic artwork showing an army of skeletons laying waste to a desolate landscape, evoking the apocalyptic horsemen.
- ***The Course of Empire: Destruction* (Thomas Cole):** The penultimate painting in Cole's series, illustrating the violent collapse of a grand civilization watched over by a giant warrior statue.
- ***The Great Day of His Wrath* (John Martin):** Part of a last-judgment triptych inspired by the industrial Black Country, depicting falling mountains and glowing lava under a dark sky.
- ***Der Krieg* (Otto Dix):** A stark print portfolio capturing the nightmarish reality and destruction of World War I from a German soldier's perspective.
- ***Apocalypse 42* (Viktor Schreckengost):** A terracotta sculpture created during World War II depicting Hitler, Mussolini, and Hirohito riding alongside a terrified horse.
- ***The Last of Us* Concept Art (Saby Menyhei):** Environmental art for the post-apocalyptic television series illustrating overgrown, ruined skyscrapers reclaimed by

nature.

- **"Where Are We Going" (Kevin Sherwood & James McCawley):** A haunting song featured in the "Mob of the Dead" zombie map within *Call of Duty: Black Ops II*.
- ***Quartet for the End of Time: Abyss of the Birds* (Olivier Messiaen):** The third movement of a piece composed while Messiaen was held in a German prisoner-of-war camp, written for solo clarinet.
- **"Electric Funeral" (Black Sabbath):** A heavy metal track delivering a bleak warning regarding the destructive potential of nuclear warfare.
- **"It's the End of the World as We Know It" (R.E.M.):** A high-tempo song featuring rapid-fire lyricism filled with disparate cultural and historical references.

Environmental & Global Risk Factors

- **Chlorophyll & Ecosystem Collapse:** Sudden spikes in chlorophyll levels serve as an early warning indicator for lake ecosystem collapses, demonstrated when introducing top predators alters fish behavior and allows phytoplankton to bloom.
- **Increased Plane Turbulence:** Severe turbulence—exerting forces over 1.5g—is growing more frequent on major flight paths due to climate change, costing airlines millions annually in maintenance and altered flight routes.
- **Stardust Solutions:** A solar geoengineering company aiming to counter global warming by releasing reflective particles into the atmosphere to redirect sunlight back into space.

There is A Draft in here

Lin-Manuel Miranda, This one's mine, 2018

From blueprint to Broadway brilliance

This demo, written for Hamilton but eventually cut as a Hamildrop, features Lin Manuel Miranda's signature composer vocals; he's focusing on the rhythm and rhymes rather than melodies.

What changed: In the demo the piano is purely functional, keeping the beat. In a finished cast recording the orchestration swells to match the character's emotional stakes.

The finished factor: A demo often sounds like someone explaining a song. The finished product sounds like a character living it. The arrangement adds breath, knowing when to pull back and when to explode.

Sophie Anne Caruso's dead mom 2019

From vulnerability to power

The demo for this Beetlejuice standout is already quite strong, as Caruso originated the role. However, the demo feels more like a rehearsal room recording.

What changed: The finished cast recording features a "wall of sound" approach. The electric guitars are grittier, and the drums are snappier, leaning into the pop punk energy required for the character.

The Finished factor: In the final version, the vocal is compressed and EQ-ed to sit right at the front of the mix. The background vocals are layered perfectly to create a sense of scale that a solo demo lacks.

EJAE Golden 2025

From viral spark to sonic depth, Ejae's process often involves sharing snippets and work-in-progress versions. The golden demos highlight the modern transition from a vibe to a structured song.

What changed: The early demos usually focus on the hook that golden earworm. The finished product builds a narrative around it. The production likely evolved from a simple loop to a dynamic arrangement with transitions that keep the listeners engaged for three minutes.

The finished Factor: Modern finished tracks heavily rely on "ear candy" tiny textures, vocal chops, and bass frequencies that are often absent in the voice memo stage.

Strawberry Fields Forever 1966

In 1966, the Beatles were exhausted by fame, and they decided to stop concerts and spend all their time in the recording studio. The song started very simply. Early recordings show John Lennon alone with an acoustic guitar. It

sounded like a sad, simple folk song.

- The Transformation: The Beatles didn't want it to stay simple. They used the studio to turn it into something "psychedelic"—meaning it sounds like a dream or a strange vision.

- The Final Result: They combined two different versions of the song, added backward-recorded drums, and used orchestral instruments to create a "soundscape." That feels like stepping into someone's memory.

Mona Lisa Mona lisa isleworth

Portrait of Lisa Gherardini by Da Vinci.

**Isleworth: The earlier versions background was crude and undetailed

Acquired by Hugh Blaker

Got the name Isleworth because Blakers studio was there

Mixed opinions whether it was Da Vinci's work or not

Bedroom in Arles Van gogh

Made 3 almost identical versions

1st one used the most saturated colours, was destroyed in a flood while he was in a hospital in Arles

He then made 2 copies of the same work

one was smaller and more detailed meant for his family

The other was larger

Van Gogh was referring to Japan through the various colours and simple interior

Despair and The scream Edvard munch

Despair is the 1st version of scream

expresses the feelings of anxiety he had when he went on a walk

Depicts a man standing on a bridge in front of a cliff at sunset

Takes place at Kristiania fjord

The scream

Everything becomes more distorted and lower resolution than Despair

While the setting looks similar the man is replaced with a mysterious figure

even has its own emoji

Dance Henri Matisse

was intended to hang on the landing of a staircase

Its over simplification was attacked as willfully crude

The 2nd version adds a lot more detail and more saturated colors.

Aladdin Count on me 1994

Context: This song was intended for the original animated film. It features Aladdin singing to his three friends Babaka Omar and Kassim.

Why it was left out:

Character streamlining: Early script drafts feature a large ensemble of human sidekicks. To simplify the narrative and focus on the lead's journey, the directors replaced the trio with a single non verbal companion Abu the monkey.

Loss of purpose: Without the buddies the song lost its narrative anchor. While the characters were eventually restored for the Broadway stage adaptation, the production opted for new numbers that better suited the energy of a live musical.

Pocahontas First to dance 1995

Context: A playful upbeat demo featuring Pocahontas and Nakoma discussing the arrival of the settlers and the potential of meeting them.

Why it was left out:

A shift in tone: Disney initially developed Pocahontas with a lighter more traditional musical comedy feel. However the studio eventually pivoted toward a sweeping epic romance with a more spiritual and serious atmosphere.

The I want song replacement: Creators felt that first to dance was too bubbly for the new direction. It was ultimately cut in favor of “Just around The Riverbend,” which better established Pocahontas’s internal conflict and soulful nature.

Book of Mormon “Family home evening” 2009

A Mormon is a member of the church of Jesus Christ of latter day Saints. While the world often use the nickname, the church prefers the term Latter day saint. They are a Christian group with different theological views than the Catholic Church; notably they study the Bible along with three other books of scripture, including the Book of Mormon.

Context: Family home Evening an early opening that depicted a traditional Mormon family night, introducing the perfect LDS lifestyle.

Why it was left out:

The Efficiency of “Hello!”: The creative team realized that the most effective way to introduce the world wasn’t through a domestic scene, but through the Missionary Training Center.

Pacing and Impact: While “Family Home Evening” provided great world building, it lacked the immediate momentum and comedic punch of Hello successfully introducing the characters and the theology, and the humor in one rapid fire sequence.

Dear Evan Hansen In the bedroom down the hall 2017

Context: Dear Van Hansen is a contemporary musical exploring the complexities of mental health, the impact of social media and the universal need for connection. A heartbreaking demo from the show, In the bedroom down the hall is a duet between two mothers reflecting on their sons' empty rooms. It captures a shared grief: one mothers son is physically gone while the other is physically present but emotionally distant.

Why it was left out:

Perspective shift: The narrative of Dear Evan Hansen centers almost entirely on the teenage experience. While this song is hauntingly beautiful, the creators felt that having two adult characters lead a long somber ballad in the second act stalled the momentum. At that point in the show, the audience is waiting for the ticking clock of Evans lies to finally explode.

Protecting the Climax: Sometimes a great song must be sacrificed for the sake of the larger story. By removing this ballad, the creators ensured It was a strategic cut made to maintain

structural tension, preventing the pace from sagging right when stakes needed to be at their tightest. the show's final emotional punch remained powerful.

Were all in this together

Famous Long-Term Construction Projects

La Sagrada Família

La Sagrada Família is a Catholic basilica located in Barcelona, Spain, and was designed by the renowned architect Antoni Gaudí. It is currently the oldest active construction project in the world. When asked why the project was taking so long, Gaudí famously replied, “My client is not in a hurry,” referring to God. The stained-glass windows are arranged so that brighter colours are placed lower while more translucent glass is positioned higher, making the interior appear cooler in the morning and warmer in the evening. Gaudí’s design for both light and structure was heavily inspired by the patterns created by a kaleidoscope. Construction was severely disrupted during the Spanish Civil War. Once completed, the basilica is expected to become the tallest church building in the world. Current estimates place its completion sometime between 2026 and 2030.

Nusantara

Nusantara is a planned sustainable smart city in Indonesia that is intended to become the country’s new capital. The project aims to redistribute economic growth beyond the island of Java, and its location was chosen because it lies close to the geographic centre of Indonesia. The relocation was formally proposed during the presidency of Joko Widodo. After a change in political leadership, progress slowed considerably as the government’s priorities shifted toward food security, energy infrastructure, and development closer to existing population centres. Although many people believe the project has been abandoned, it has never been officially cancelled. Historically, many attempts to relocate national capitals have struggled to achieve their intended goals.

Ryugyong Hotel

The Ryugyong Hotel in North Korea is an unfinished pyramid-shaped skyscraper designed to function as a hotel and mixed-use building. Construction stopped after the collapse of the Soviet Union triggered a severe economic crisis in North Korea. The hotel was later scheduled to open in 2012, but this deadline was missed. In 2018, LED screens were installed on its exterior to display state propaganda. Had it opened on schedule, it would have been the tallest hotel in the world. For many years, it also held the distinction of being the world's tallest unoccupied building.

Wonderland Amusement Park

Wonderland Amusement Park in China was originally planned to become the largest amusement park in Asia. Construction came to a halt in 1998 because of financial difficulties. Although several attempts were made to restart the project, none succeeded. The abandoned structures were eventually demolished in 2013, and a commercial shopping centre was later built on the site.

Berlin Brandenburg Airport

Berlin originally operated three airports—Tegel, Tempelhof, and Schönefeld—which proved inefficient for serving the city. To replace them, Berlin Brandenburg Airport was built next to the existing Schönefeld Airport. The airport was originally scheduled to open in October 2011, but serious failures in the fire safety system and backup generators delayed its launch. Numerous other technical problems were later discovered, causing repeated postponements. After almost a decade of delays and extensive repairs, the airport finally opened in 2020.

Milan Cathedral

Milan Cathedral took nearly six centuries to complete. Construction began in 1386 under the patronage of Milan's rulers and involved many architects and artists over the centuries. Napoleon later ordered and funded the completion of the cathedral's façade. Although the cathedral was officially

completed in 1965, some marble blocks remain uncarved so that future statues can still be added.

Crazy Horse Memorial

The Crazy Horse Memorial in South Dakota was commissioned to honour the famous Native American leader Crazy Horse. Sculptor Korczak Ziolkowski devoted his entire life to carving the monument into a granite mountain. It is intended to become the world's largest monument, yet after more than 70 years of work it is still far from complete. Because no verified photographs of Crazy Horse exist, the face of the sculpture is symbolic rather than historically accurate. The memorial stands only 17 miles from Mount Rushmore, and construction progresses slowly with a relatively small workforce.

Project Management

Gantt Chart

The Gantt chart was developed by engineer Henry Gantt to improve project management. It organises tasks into horizontal bar charts, making project scheduling, progress tracking, and resource allocation much easier. The construction of the Hoover Dam relied heavily on Gantt charts to coordinate its enormous workforce and resources.

The Iron Triangle

The Iron Triangle is a project management model stating that every project is constrained by three factors: scope, time, and cost. According to the model, it is usually possible to optimise only two of these at once. For example, completing a project more quickly generally increases its overall cost.

Kanban

Kanban is a visual scheduling system created by industrial engineer Taiichi Ohno to improve manufacturing efficiency. The name comes from the Japanese word for the cards originally used to track production. Kanban aims to reduce excess inventory by limiting the number of active tasks in each stage of production. Work items are represented as cards that move across columns as progress is made, and new tasks are started only when capacity becomes available.

Harmonogram

A harmonogram is the Polish term for a structured project schedule or timeline. Like a Gantt chart, it displays tasks as horizontal bars to show their duration and important milestones.

Dependency

A dependency describes the logical relationship between project tasks. For example, a start-to-start dependency means one task must begin before another task can start.

Productivity and Management Principles

Hawthorne Effect

The Hawthorne Effect is the tendency for people to change their behaviour simply because they know they are being observed. The term originated from productivity studies conducted at the Hawthorne Works factory, where employees manufactured electrical relays.

Hofstadter's Law

Hofstadter's Law states that complex tasks always take longer than expected, even when the delay is already taken into account. Early

predictions about how quickly computers would defeat humans at chess are often used as an example.

The 90–90 Rule

The 90–90 Rule humorously states that the first 90% of a software project takes 90% of the total development time, while the remaining 10% also takes the other 90%.

Conway's Law

Conway's Law states that organisations tend to design software systems that reflect their own internal communication structures.

Putt's Law

Putt's Law suggests that technology is dominated by two groups of people: those who understand what they do not manage and those who manage what they do not understand.

Illich's Law

Illich's Law proposes that productivity declines after a certain amount of continuous work and may eventually become negative. This point is generally reached after about 90 minutes to two hours of uninterrupted effort.

Laborit's Law

Laborit's Law, also known as the Law of Least Effort, suggests that people naturally prefer easier tasks with immediate rewards rather than difficult tasks that produce greater long-term benefits.

Brooks' Law

Brooks' Law states that adding more people to a delayed software project will usually make it even later because new team members require training and increase communication complexity.

Pareto Principle

The Pareto Principle, also known as the 80–20 Rule, states that roughly 80% of results come from 20% of the causes or effort. It was named after the Italian economist Vilfredo Pareto, who observed a similar pattern in wealth distribution.

Six Sigma

Six Sigma is a data-driven methodology designed to achieve extremely high levels of quality by reducing defects and process variation. It relies heavily on statistical analysis to improve business performance.

Waterfall Method

The Waterfall Method is a linear and sequential approach to project management in which each phase must be fully completed before the next phase begins. This makes projects predictable but also makes changes difficult once development has started.

Command and Control

Command and Control is a top-down management style in which decisions are made by a central authority. It emphasises discipline, consistency, and predictability rather than employee independence.

Toyota Production System

The Toyota Production System is a lean manufacturing philosophy focused on eliminating waste and improving efficiency. It combines Just-in-Time

production with Jidoka, a principle that allows workers to stop the production line immediately whenever a defect is discovered.

Chaos Engineering

Chaos Engineering is the practice of deliberately introducing controlled failures into computer systems to identify weaknesses. By testing systems under stressful conditions, organisations can improve reliability and resilience before real failures occur.

Agile Manifesto

The Agile Manifesto is a set of principles for software development that values individuals and interactions, working software, customer collaboration, and responding to change over rigid planning and excessive documentation. Its goal is to help teams adapt quickly to changing requirements.

Sashimi Model

The Sashimi Model is a variation of the Waterfall approach that allows project phases to overlap. This overlap enables work on the next stage to begin before the previous stage has been completely finished, creating a smoother project timeline.

Phoenix Project Philosophy

The Phoenix Project philosophy, often referred to as the “Three Ways,” teaches that successful organisations maximise the flow of work, create rapid feedback loops, and promote continual learning and experimentation. Its goal is to transform IT from a bottleneck into a strategic advantage for the business.

Where the sidewalk starts

Temple street Parking garage New Haven

Designed by Paul Rudolph in 1962

700 feet long

6 stories high

Lego parking house Billund Denmark

By architecture studio CEBRA Covered with a metal sheet with lego stud sized holes You can attach lego bricks to its exterior

Parking des Célestins Lyon France

By Michel Targe, Jean Michel Willmore and Daniel Buren Buren's work "Sens dessus dessous" (Upside down) can be seen through a square periscope

Michigan Theater Detroit

It was closing down due to lack of profits where the lack of parking was a big factor. The theater has not been converted into a parking lot.

Garagenatiler Herdern Switzerland

A weird Car park that only has space for 8 cars

9th Avenue Parkade Calgary Canada

Is a Parking lot that can be transformed into a 600 person office or residential building can fit up to 510 cars In the shape of a long oval and ramps to higher floors have a very low level of inclination Making it easy to convert

Alfred Speer

Patented the idea of moving sidewalks The idea was to transport pedestrians along 3 belts where each belt has a different speed

Hildebrands

German chocolate manufacturer that made cards depicting moving sidewalks in the year 2000

Snot Houses

Houses in which the garage pokes out like a big snout. For most houses in the US the garage is the most prominent element is a Utility first design where use is more important than aesthetic.

Frank Lloyd wright

Wanted to include garages within the house and not as a separate structure which was called the carport. Frank Loved spiral ramps around the buildings he was famous for designing falling water and the Guggenheim museum

Robie house Designed by Frank Lloyd Wright for Frederick Robie . Located in Chicago is a Prairie style build. Last of wrights true prairie houses

Grandpark

Located In downtown LA. Transformed from a massive parking lot into a community gathering place

Monkey see Monkey prototype

THE PROTOTYPING CYCLE

1. Ideation and conceptualisation. This phase is all about brainstorming and defining what problem product will solve.
2. Creating low fidelity prototypes
3. Developing high fidelity prototypes
4. User testing and feedback
5. Iteration and Refinement
6. Final design and Handoff

Types of Prototyping

Sketches: Fast freehand drawings that are low fidelity and their purpose is to get a rough idea about the model

Storyboarding: Simple illustrations displayed in sequence. Used for prototyping movies and other motion graphics.

Was developed at Walt Disney Productions and it has a similar format to comics.

Paper Prototypes: Hand sketched or printed 2d versions of the final product. These prototypes are low fidelity

Low fidelity prototypes: Used in early stages of prototyping. Usually varies significantly from the final product

Inexpensive method.

High Fidelity prototypes: Final stages of prototyping. Very similar to the final product. Usually is a working model

Wireframing: Low fidelity Blueprint of a digital product. Shows the skeleton of a website/application. Used for content placement and testing functionality.

Mockup: A model of the final product. Is high fidelity. Used for final testing and can be full sized

Proof of concept: Small scale model proving feasibility of an idea. Helps to avoid pouring of resources into something impossible.

User testing: A high fidelity prototype of an idea given to the target consumers to test.

Minimum viable product: A version of a product with absolute minimum features that can be tested by users

Minimum marketable feature: Absolute smallest feature of a product that gives value to a user

Sarah Boone

- African American dressmaker
- Awarded a patent for her improved ironing board
- Until then dressmakers were ironing on a wood plank placed across two chairs
- Fine for a wide skirt but not for tight fitted material
- Created a narrower curved board that slipped into sleeves

Vibe Coding

- Using AI to generate code
- They get the codework using ai then refine it
- Helps in fast prototyping
- Makes debugging increasingly difficult
- And overall code quality drop

Mechanical Turk

- By a Hungarian man called Wolfgang von Kempelen
- Automatic chess playing machine from 1770
- Consisted of a mechanical man who sat at a wooden cabinet that came with a chessboard
- When Napoleon tried to cheat by making an illegal move it moved the piece back
- Was human controlled and has fake gears to look like a machine
- Impressed Charles Babbage and Benjamin Franklin
- People suspected that he was using tiny humans to control it

Wizard of Oz Technique

- Interactions with the prototype are done by invisible “wizards” working behind the screen
- People review it on the assumption that it fully works

Tesla bot

- The prototype turned out to be a man dancing in a costume
- Instead of an actual prototype of the humanoid

Tesla's Optimus humanoid

- Stops working when the teleoperator takes off his headset

Amazons Just walk out

- Cashierless stores that use AI checkout
- You could grab anything and the AI would charge it to your Amazon account
- Turned out to be a 1000 human video reviewers in India

- Replaced it with shopping carts with a scanning system

Fireflies ai

Claimed to be an AI meeting notetaking software
Was valued at 1 billion dollars
Turns out it was one of the founders manually taking notes
They would join the meet as Fred from Fireflies.ai and listen in to make notes

Early Prototypes of popular gadgets

- **Nintendo PlayStation (1988–1991):** A collaborative SNES CD-ROM add-on/standalone console by Nintendo and Sony before their licensing deal collapsed.
- **Game Boy Color "Atlantis":** A 32-bit ARM RISC processor prototype developed between 1995 and 1996 prior to the final handheld release.
- **Apple I (1975–1976):** Designed by Steve Wozniak and sold for \$666.66 after Steve Jobs secured an order from a local computer shop.
- **Atari VCS / 2600 (1977):** An early design that popularized microprocessor-based hardware and ROM cartridges.
- **Super Soaker (1982):** Invented by Lonnie Johnson using PVC pipe, Plexiglass, and an empty plastic soda bottle.
- **Push-button Telephone (1941–1963):** Early Bell Labs designs used plucked mechanical reeds for key tones before solid-state electronics took over in 1963.
- **Original Xbox (1998):** Constructed by Microsoft engineers using disassembled Dell laptop hardware into a massive X-shaped chassis.
- **Apple Touchscreen Phone & Tablet (1983):** Early concepts designed by Hartmut Esslinger, featuring a touchscreen with a stylus ("Bashful" tablet concept).

The lovely and the Liminal

- The **doorway effect** describes a psychological phenomenon where passing through a doorway into a new room causes people to forget their immediate thoughts or intentions.
- This memory lapse occurs because the human brain clears short-term memory upon entering a new setting, treating the physical threshold as an event boundary.
- **Liminality** refers to a transitional state of being positioned in-between two distinct stages, environments, or phases of existence.
- **The Backrooms** originated from a 2019 4chan thread as a creepypasta centered on the eerie, unsettling aesthetic of transitional liminal spaces.
- In online lore, this environment is depicted as an infinitely vast, maze-like layout of empty rooms that a person enters by "noclipping" out of normal physical reality.
- Content creator Kane Pixels developed a popular **Backrooms web series** on YouTube exploring this concept through a narrative lens.
- The series follows Async, a fictional research institute that discovers the anomaly and attempts to systematically document and explore its interior.
- The video game **POOLS** heavily incorporates liminal space aesthetics by dropping players into a silent, surreal environment composed of interconnected, abandoned indoor pools inspired by Backrooms lore.
- **The Stanley Parable** explores themes of isolation and choice, following an office worker named Stanley who wakes up to discover that all of his coworkers have mysteriously vanished.
- The game relies on eerie, surreal landscapes, empty office corridors, and foggy non-places to build an atmosphere of psychological uncertainty.

Music About Liminality

1. Yilan Folk Song – “Diu Diu Dang Ah” (c. 1924)

- **Focus:** How Art and Music Tell the Story of Migrants

- **The Sound & Onomatopoeic Title:** "Diu Diu Dang" mimics the rhythmic sound of water dripping from a tunnel's ceiling onto the train's roof, blending with the clatter of wheels on the tracks.
- **The Context (Technological Awe):** In 1924, moving through a mountain in a metal machine was a supernatural, liminal experience for a traditional farming society transitioning into the modern era.
- **The Origin:** Taiwan's famous folk song, originating from Yilan. It was popularized during the construction of the Caoling Tunnel, which linked the isolated coast.
- **Liminality Matrix Details:** Auditory / Psychological Type. Captures sensory deprivation and the suspense of moving through a pitch-black tunnel.

2. J.M.W. Turner – *Rain, Steam, and Speed* (1844)

- **The Scene:** A black steam locomotive charges across the railway. Ahead of the train, a tiny hare dashes, while a small boat drifts faintly below.
- **Key Themes:**
 - *Industrialization:* The unstoppable force of human engineering reshaping the world.
 - *Nature vs Machine:* A violent, beautiful collision between natural elements and technology.
 - *Pure Velocity:* The sheer speed of modern life dissolving the landscape into a physical blur.
- **The Context:** Painted during the height of the British Industrial Revolution. Turner captures the profound societal anxiety and awe surrounding rapid technological advancement. The train is viewed not just as a vehicle, but as a raw force of nature tearing through the traditional agrarian landscape.
- **Liminality Matrix Details:** Historical / Atmospheric Type. Focuses on the violent, blurry transition from an agrarian past to an industrial future.

3. William Powell Frith – *The Railway Station* (1862)

- **The Scene:** A hyper-detailed, bustling panoramic view of Paddington Station in London.
- **Key Themes:**
 - *Social Collision:* The meeting of disparate classes in a shared space.
 - *Anxiety & Drama:* Collective traveling anxiety and emotional extremes.
 - *Microcosm:* The station as a contained container of human experience.
- **The Context:** Mid-19th-century Victorian England. The massive expansion of the railway network fundamentally changed society. For the first time, people from vastly different economic backgrounds and geographic regions were forced to share the exact same physical threshold, making the station a unique theater of modern life.
- **Liminality Matrix Details:** Social / Spatial Type. Explores the station as a temporary microcosm where distinct social classes are forced together.

4. Tsukioka Yoshitoshi – *Picture of the Railroad at Takanawa* (1871)

- **The Scene:** A heavy, black steam train billows dark smoke as it chugs along a coastal stone causeway in Tokyo. White sails of traditional boats dot the blue bay in the background.
- **Key Themes:** Cultural collision, the shock of modernization, and the bridging of Eastern tradition with Western industrialism.
- **The Context:** Created during the early years of the Meiji Restoration, a period when Japan rapidly opened its borders to the West and modernized after centuries of isolation. The railway was the ultimate symbol of this new, foreign era; placing it within a traditional *ukiyo-e* style powerfully captures a nation in transition.
- **Liminality Matrix Details:** Cultural / Aesthetic Type. Highlights Japan's transition, bridging traditional culture (*ukiyo-e*) and Western modernization.

5. Claude Monet – *Gare Saint-Lazare* (1877)

- **The Scene:** The interior of a busy Parisian train station. Instead of focusing on the trains or the people, the canvas is dominated by thick, swirling clouds of blue, grey, and purple steam. The rigid, modern iron-and-glass roof of the station is visible but largely obscured and softened by the atmospheric haze.
- **Key Themes:** The ethereal nature of modern life, the dissolution of solid architecture, and the capture of a fleeting, sensory moment.
- **The Context:** Painted at the height of the French Impressionist movement. Monet was less interested in the literal engineering of the station and more fascinated by how modern machines created entirely new visual atmospheres. He convinced the stationmaster to delay the trains and pack them with coal just so he could paint the sensory experience of smoke trapping the sunlight.
- **Liminality Matrix Details:** Architectural / Sensory Type. Focuses on the dissolution of solid structures into a fleeting, hazy atmosphere of steam and light.

6. Steve McCurry – *Train Station, Agra, 1983*

- **The Scene:** A documentary photograph showing Indian railway passengers curled up and sleeping directly on the concrete platform, surrounded by their meager luggage. The lighting is cinematic, highlighting the deep exhaustion of the subjects. The everyday chaos of the station is paused in a moment of deep physical rest.
- **Key Themes:** The physical toll of transit, the suspension of daily routine, endurance, and the reality of travel as a state of waiting rather than a state of motion.
- **The Context:** Late 20th-century India. The sprawling Indian railway network is the lifeblood of the country, moving millions of people daily. McCurry's photography strips away the romanticism of travel, focusing instead on the grueling, exhausting reality of everyday citizens caught in the "in-between" of their journeys.
- **Liminality Matrix Details:** Temporal / Physical Type. Portrays the exhaustion of transit and the reality of travel as a suspended state of waiting.

7. Richard Artschwager – *Six in Four* (2015)

- **The Scene:** The interior cabs of the four massive elevators at the Whitney Museum of American Art. The walls are covered in intricate patterns made from Formica, glass, and mirrors. Each elevator has a different theme—one looks like the inside of a woven basket, another creates an optical illusion of a floating room.
- **Key Themes:** Disorientation, the surrealism of mundane transit, and forced introspection.
- **The Context:** Contemporary public art in New York City. Elevators are the ultimate liminal spaces: small, awkward boxes we stand in purely to get somewhere else, usually while ignoring our surroundings. By turning the interior into a surreal, immersive installation, Artschwager forces the rider to actively experience the few seconds of isolation before the doors open again.

Poems about Liminality

. Samuel Beckett | *Waiting for Godot*, Act 2 (1953)

- **Context & Meaning:** As a cornerstone of the **Theatre of the Absurd**, Beckett's play explores the fundamental human habit of waiting for a savior, purpose, or meaning that never arrives. Act 2 mirrors Act 1 with agonizing precision, emphasizing the repetitive, cyclical nature of time.
- **Key Dynamics in Act 2:**
 - * **The Tree:** The barren tree from Act 1 has suddenly grown four or five leaves, a mocking hint of hope or the passage of time in an otherwise static world.
 - **Memory Decay:** Vladimir remembers the past, but Estragon does not. Pozzo returns, but he is now blind, and Lucky is mute. Their sudden physical decay suggests that time is passing destructively, even though nothing is actually happening.
 - **The Ultimate Dilemma:** The act ends exactly like the first—with the boy delivering the message that Godot will not come today, but surely tomorrow. The famous final lines encapsulate the paralysis of the human condition: "*Shall we go?*" "*Yes, let's go.*" (*They do not move*).

2. Pablo Neruda | "Keeping Still" (1958)

- **Context & Meaning:** Written during a period of intense global political tension, Neruda's poem is an urgent plea for universal silence, stillness, and introspection. It suggests that humanity's constant, frantic motion is precisely what drives us toward self-destruction.
- **Key Themes:**
 - **The Power of Silence:** Neruda asks everyone on Earth to count to twelve and freeze for a single second. In this "exotic moment," the normal structures of violence and exploitation—fishermen hurting whales, men gathering salt until their hands bleed—would cease.

- **"Clean Clothes":** He envisions a world where enemies put on clean clothes and walk together under the shade of trees, doing nothing.
- **Stillness $\neq$ Death:** Neruda carefully clarifies that he is not advocating for total inactivity or death. Instead, he argues that a temporary pause would allow us to break our existential sadness and finally understand ourselves.

3. Philippe Orreindy | *I'll Wait for the Next One...* (*J'attendrai le suivant*) (2002)

- **Context & Meaning:** This French short film is a devastating, bittersweet look at urban loneliness, desperation, and the cruelty of false hope. It centers on a plain, isolated woman riding the Paris Metro.
- **Key Plot & Mechanics:**
 - * A man boards the train and makes a passionate, theatrical speech to the entire car. He claims he is looking for love, is tired of being alone, and asks any woman who is interested to get off at the next station with him.
 - The protagonist is deeply moved by his vulnerability and decides to take a leap of faith. She steps off the train, only to watch the man immediately begin the exact same speech to a new crowd on the platform. It was all a street-performer's act.
- **Theme:** The film highlights the invisible walls between people in crowded modern cities and the tragic vulnerability of someone desperate for human connection.

4. Marcel Aym   | "The Man Who Could Walk through Walls" (1941)

- **Context & Meaning:** Written during the German occupation of France, this satirical, magical realist short story uses a fantastic premise to comment on bureaucracy, freedom, and the human tendency to self-sabotage.
- **Key Plot & Elements:**
 - **Dutilleul**, an ordinary, unremarkable clerk, suddenly discovers he can walk through walls. Initially, he uses this gift modestly. However, when a tyrannical new boss makes his life miserable, Dutilleul uses his power to drive the boss insane, pull off bank robberies, and mock the police.
 - **The Twist:** He eventually falls in love with a married woman. After a romantic night, he takes some pills to cure a headache—pills that, unknown to him, actually cure his wall-walking ability. As he leaves her house, he gets permanently stuck mid-way inside a wall.
- **Theme:** The story serves as an allegory for resistance against confinement, but ends with a darkly ironic twist: the moment Dutilleul finds true passion, he is permanently immobilized by his own carelessness.

5. Amor Towles | *A Gentleman in Moscow* (2016 / Excerpted Context)

- **Context & Meaning:** The novel follows Count Alexander Rostov, an unrepentant aristocrat sentenced by a Bolshevik tribunal in 1922 to spend the rest of his life inside the luxurious Hotel Metropol. If he steps outside, he will be shot.
- **Key Themes in the Confinement:**
 - **Internal Freedom:** Rather than letting his tiny attic room become a prison, the Count decides that *"if a man does not master his circumstances then he is bound to be mastered by them."* He actively chooses to live a life of purpose, elegance, and deep relationships within his confined borders.
 - **A Microcosm of History:** Through the walls and windows of the hotel, Towles shows decades of tumultuous Soviet history playing out. The hotel becomes a miniature world where culture, manners, and human decency are preserved despite the chaos outside.

Art pieces about liminality

1. Paul Gauguin | *Where Do We Come From? What Are We? Where Are We Going?* (1897–98)

- **Context & Meaning:** Painted in Tahiti during a period of severe personal crisis, this massive canvas is Gauguin's ultimate artistic and philosophical testament. Read from **right to left**, it represents the cycle of human life.
- **Key Elements:**
 - * **Right (Where do we come from?):** A sleeping infant and young women represent the beginning of life and innocence.
 - **Center (What are we?):** Figures going about daily life, with a central figure plucking fruit, symbolizing the temptation or the experiencing of existence.
 - **Left (Where are we going?):** An old woman approaching death, paired with a strange white bird representing the futility of words or the unknown afterlife.
- **Style:** Post-Impressionist/Synthetist. Gauguin uses vivid, non-naturalistic colors and flat planes to evoke a dreamlike, spiritual atmosphere rather than a realistic scene.

2. René Magritte | *The Treachery of Images* (1929)

- **Context & Meaning:** This masterpiece of Surrealism is a profound philosophical joke about the nature of representation. By painting a hyper-realistic pipe and writing beneath it *"Ceci n'est pas une pipe"* (This is not a pipe), Magritte forces the viewer to confront a literal truth: it *isn't* a pipe; it is a *representation* (a painting) of one.
- **Key Elements:** The crisp, advertising-style execution strips away emotional pretense, turning the artwork into a linguistic and visual puzzle.

- **Style:** Surrealist/Conceptual. It challenges the historical assumption that art's primary goal is to perfectly mimic reality, highlighting the gap between language, objects, and images.

3. Salvador Dalí | *The Persistence of Memory* (1931)

- **Context & Meaning:** Perhaps the most famous Surrealist painting in history, this work explores the fluidity of time and the weight of the subconscious mind. Dalí claimed the idea of the melting clocks came from watching Camembert cheese melt in the sun.
- **Key Elements:**
 - **Melting Clocks:** Symbolize the breakdown of fixed, rational time in the dream world.
 - **The Biomorphlic Creature:** The distorted face in the center is a self-portrait of Dalí, sleeping, suggesting the entire landscape exists within his dream.
 - **The Ants:** Gathering on a closed pocket watch, they represent decay, mortality, and anxiety.
- **Style:** Surrealism. Dalí uses a hyper-realistic, academic painting technique—which he called "hand-painted dream photographs"—to make the impossible feel disturbingly real.

4. Andrew Wyeth | *Christina's World* (1948)

- **Context & Meaning:** At first glance, this looks like a peaceful pastoral scene, but it quickly shifts into something deeply tense. The subject is Anna Christina Olson, Wyeth's neighbor in Maine, who had a degenerative muscle disease (likely polio or Charcot-Marie-Tooth disease) and refused to use a wheelchair, preferring to crawl.
- **Key Elements:**
 - **The Pose:** Christina is viewed from behind, her body taut with effort as she looks toward her distant home.
 - **The Scale:** The vast, dry field emphasizes her isolation and the immense physical challenge of her journey.
- **Style:** Regionalism / Magic Realism. Wyeth used egg tempera, a meticulous medium that allowed him to paint individual blades of grass, creating a sharp, hyper-focused clarity that amplifies the psychological tension.

5. M. C. Escher | *Ascending and Descending* (1960)

- **Context & Meaning:** Based on the "Penrose stairs" (an impossible mathematical object), this lithograph is a brilliant exploration of paradox, infinity, and human futility.
- **Key Elements:** * **The Monks:** Two lines of robed figures walk endlessly around the roof of a monastery. One line climbs indefinitely, while the other descends, yet they both remain on the exact same level.

- **The Outcasts:** A few figures sit apart from the stairs, refusing to engage in the endless, meaningless ritual.
- **Style:** Mathematical Art / Op Art. Escher uses precise architectural perspective to trick the human brain into accepting a structure that cannot physically exist in three-dimensional space.

6. Gregory Crewdson | *Beneath the Roses* (2004)

- **Context & Meaning:** This is a massive photographic series rather than a single image, exploring the dark, anxious underbelly of small-town American suburbs. Crewdson's work deals with isolation, psychological trauma, and the uncanny hidden behind the "American Dream."
- **Key Elements:** The images feature ordinary people caught in frozen, inexplicable moments of despair or detachment (e.g., a woman standing in a flooded living room, or a man staring blankly out a window).
- **Style:** Cinematic Photography. Crewdson operates like a film director, using huge crews, elaborate lighting rigs, and Hollywood-level production budgets to create hyper-detailed, melancholic tableaux that feel like stills from a movie that doesn't exist.

7. Camille Seaman | *The Distant is Imminent* (2020)

- **Context & Meaning:** Part of a powerful contemporary climate-focused project, Seaman's work bridges the gap between urban human life and the melting polar ice caps. By projecting massive, tragic images of calving icebergs onto the architecture of major cities, she visualizes the immediate threat of climate change.
- **Key Elements:** The juxtaposition of ancient, pristine, melting ice against the gritty, temporary monuments of human civilization. It serves as a stark reminder that what happens "far away" will soon reshape our immediate doorsteps.
- **Style:** Environmental Conceptual Photography / Public Installation. Seaman uses high-contrast, emotionally charged imagery to evoke awe, grief, and a sense of urgent responsibility.

Music about anxiety

2. Brian Eno | *Ambient 1: Music for Airports* (1979)

- **The Origin:** Eno conceived the idea while stuck in a "dreary" German airport. He wanted to create

music that would defuse the specific anxiety associated with air travel.

- **The Concept:** This was the first album to use the term "Ambient." Eno's goal was to create music that

was "as ignorable as it is interesting."

- The Technique: He used tape loops of different lengths that overlap at irregular intervals. Because the

loops never align the same way twice, the music feels "alive" and non-repetitive, preventing the "trapped" feeling of traditional background music (Muzak).

3. Hiroshi Yoshimura | Music for Nine Postcards (1982)

- The Origin: A pioneer of Japanese "kankyō ongaku" (environmental music), Yoshimura recorded

this album at home using a Fender Rhodes and a keyboard.

- The Inspiration: He was inspired by the view from his window—the movement of clouds, summer

shadows, and rain. He originally sent a cassette of the music to the Hara Museum of

Contemporary Art to be played specifically for visitors looking out at the museum's courtyard.

- The Vibe: Unlike Eno's "public" ambient, this is deeply intimate. It treats the home or a gallery as

a space where the "outside" (nature) and "inside" (the listener) meet.

4. Max Richter | Sleep (2015)

- The Origin: An 8-hour composition designed to be listened to while the audience is actually sleeping.

- The Goal: It is a protest against the fast pace of modern life, offering a massive, slow landscape

for the subconscious mind.

- The Liminality: It targets the most profound liminal state—the threshold between consciousness

and the dream state.

Going pains

Cultural and Traditional Ceremonies

- **Sweet Sixteen:** A traditional American and Canadian celebration for a teenager's 16th birthday that historically marked entry into society, though today it is usually celebrated as a milestone birthday party with family and friends.
- **Fiesta de Quinceañera:** A traditional celebration in many Latin American cultures marking a girl's 15th birthday, symbolizing the transition from childhood to young womanhood through a religious mass, a formal court of young friends, a specific gown, and a waltz.
- **Guan Li and Ji Li:** Ancient Chinese coming-of-age rituals where Guan Li was performed for young men at age 20 (receiving an adult cap and new name) and Ji Li was for young women at age 15 (involving a hair-pinning ceremony), both representing the embrace of adult responsibilities and Confucian virtues.
- **Seijin no Hi (Coming of Age Day):** A Japanese national holiday held on the second Monday of January that honors everyone who turned of age during the past year, with young adults dressing in traditional kimonos or formal suits to attend civic ceremonies.
- **Debuts:** A popular coming-of-age celebration in Filipino culture for a young woman's 18th birthday, featuring formal attire and traditional rituals such as the "18 Roses" dance with significant men in her life and the "18 Candles" speeches by close friends and family.

Religious and Spiritual Transitions

- **Bar and Bat Mitzvah:** A Jewish tradition where boys at age 13 (Bar Mitzvah) and girls at age 12 or 13 (Bat Mitzvah) reach religious maturity, becoming accountable for their actions under Jewish law and often leading a synagogue service as part of the celebration.
- **Khatam Al-Quran:** An Islamic tradition marking the successful completion of reading the entire Quran in Arabic from start to finish, celebrated as a major educational and spiritual milestone with family feasts and community prayers.
- **Vision Quests:** A sacred spiritual tradition in many indigenous North American cultures where a young person spends several days alone in nature, fasting and praying to connect with the spiritual world and discover their purpose as they transition into

adulthood.

- **Walkabout:** A traditional rite of passage for male Australian Aboriginal adolescents involving a prolonged period living alone in the wilderness, tracing ancestral pathways to demonstrate self-sufficiency, endurance, and connection to the land.

Modern and Legal Milestones

- **High School Graduation:** A modern educational rite of passage that represents the successful completion of mandatory schooling and entry into higher education or the workforce.
- **Driver's License:** A practical milestone representing personal freedom and mobility, alongside the trust and responsibility required to operate a motor vehicle.
- **First Paycheck:** A milestone of financial independence that marks a person's entry into the workforce, teaching the value of labor and the responsibility of money management.
- **National Service:** A mandatory period of military or civil service required by law in many countries upon reaching adulthood, serving as a collective rite of passage focused on civic duty and discipline.
- **Voting:** A key legal milestone in democratic societies that marks a person's official recognition as a participating political member with a voice in choosing leadership.
- **Moving Out:** The physical and emotional transition of leaving the childhood family home to live independently for college or work, widely viewed as a final practical step into full adulthood.

Works about teenage hood

We real cool by Gwendolyn brooks 1963

Context and meaning: This short rhythmic poem is a gritty but sympathetic look at rebellious youth peer pressure and the hidden vulnerability of teenagers trying too hard to look tough. It centers on seven young guys hanging out at a pool hall.

Key plot and mechanics: The boys boast about their rebellious lifestyle skipping school staying out late drinking and living fast. But the final line drops a sudden heavy reality check: We die soon. their pride is immediately cut short by the tragic truth of where their choices are leading.

Theme: Highlights how young people use performative rebellion and a pack mentality to hide their insecurity, showing how easily youth can be cut short when trying to act too cool.

The Medicine bag by Virginia Driving Hawk Sneve 1975

Context and meaning: This short story is a touching look at cultural identity the embarrassment teenagers often feel around family and what it truly means to grow up and honor your roots.

Key plot and mechanics: Martin loves bragging about his Great-grandfather's Lakota heritage to his friends but when the old man actually shows up to visit Martin is deeply embarrassed by his traditional appearance however when his grandfather passes down a sacred medicine bag and explains its history Martin's shame turns into deep respect he proudly accepts the heirloom

Theme: The story highlights the shift from shallow teenage peer pressure to mature adult appreciation for family history showing that growing up means carrying your heritage forward.

"In Mrs. Tilscher's Class" by Carol Ann Duffy (1990)

Context & Meaning: This poem is a nostalgic but jarring look at the exact moment childhood innocence ends and the confusing world of adolescence begins.

Key Plot & Mechanics: It starts in a colorful, safe, and magical primary school classroom where everything feels perfect. But as the school year ends and summer approaches, a classmate explains how babies are made, the air grows "feverish," and the main character suddenly outgrows the classroom, stepping out into a thunderstorm.

Theme: The poem highlights the inevitable, uncomfortable transition of puberty, where the safe boundaries of childhood melt away whether you are ready for it or not.

Where Do We Go From Here?" by Joss Whedon (2001)

Context & Meaning: This musical ensemble song captures the overwhelming anxiety, confusion, and emotional hangover that happens right after a major life-changing event.

Key Plot & Mechanics: After a series of intense truths and secrets are forcefully exposed, a group of young adults stand together in a state of emotional limbo. Instead of celebrating their survival, they repeatedly sing the question, "*Where do we go from here?*" because the comfortable status quo is gone.

Theme: The song highlights the paralysis of young adulthood—the terrifying moment when the safety net is gone, the future is a blank slate, and you realize nobody has a roadmap for what comes next.

St. Lucy's Home for Girls Raised by Wolves" by Karen Russell (2006)

Context & Meaning: This surreal short story is a dark, allegorical look at the painful pressure to conform, assimilate, and grow into what society considers a "proper adult."

Key Plot & Mechanics: A pack of human girls who were raised by wolves are taken to a strict boarding school run by nuns to be "civilized." They go through painful stages of learning to walk, talk, and behave like humans. By the end, the main character successfully blends in, but when she returns home, she realizes she is too human for her family but will always feel like a wolf to society.

Theme: The story highlights the tragic cost of growing up and fitting in, showing how adapting to society's expectations often means killing off your true, natural self.

"Lost Boy" by Ruth B (2015)

Context & Meaning: This piano ballad is a bittersweet look at loneliness, isolation, and using childhood fantasy as a coping mechanism to escape a harsh reality.

Key Plot & Mechanics: Using the lore of *Peter Pan*, the singer describes feeling completely alone and lost in the real world until Peter Pan appears and takes them to Neverland. There, they find friendship with the Lost Boys and a sense of belonging, choosing to stay in a world where nobody ever grows up.

Theme: The song highlights the deep desire to retreat into childhood imagination and regression when the real world feels too overwhelming or lonely to face.

"Let Me Begin Again" by Major Jackson (2021)

Context & Meaning: This contemporary poem is an introspective look at the adult desire for renewal, fresh starts, and the realization that growing up is a lifelong process.

Key Plot & Mechanics: Rather than focusing on a teenager transitioning into an adult, the speaker reflects on the ongoing human desire to wipe the slate clean. The poem focuses on the beauty of shedding old versions of yourself, embracing mistakes, and consciously choosing to start over.

Theme: The poem highlights the idea that maturity isn't a final destination you reach and stop; instead, true adulthood is the ongoing, lifelong resilience required to constantly reinvent yourself.

"A stone I died" by Rumi (c. 13th century)

Context & Meaning: This spiritual, mystical poem looks at existence through the lens of cosmic evolution and reincarnation. It views death not as a tragic conclusion, but as a necessary stepping stone to a higher state of being.

Key Plot & Mechanics: The speaker traces their soul's journey upward through different forms of existence—dying as a stone to become a plant, dying as a plant to become an animal, and dying as an animal to become a human. The poem concludes by looking forward to the next inevitable death, which will transform them into an angel, and eventually reunite them with the divine.

Theme: The poem highlights death as a natural, progressive vehicle for spiritual growth, showing that everything in the universe is constantly evolving into something greater.

“Death, Be Not Proud” by John Donne (1633)

Context & Meaning: This intense Holy Sonnet is a direct, defiant confrontation with the concept of mortality from a Christian perspective, stripping away the fearsome reputation that death holds over humanity.

Key Plot & Mechanics: The speaker talks directly to Death as if it were a person, mocking its supposedly massive power. He argues that death is actually just a short sleep, completely dependent on external forces like fate, chance, kings, and desperate men. The poem ends with a powerful paradox: once humanity wakes up into eternal life, *“Death, thou shalt die.”*

Theme: The poem highlights the ultimate powerlessness of mortality, showing that faith and eternity can render the terrifying concept of death completely obsolete.

“Ulysses” by Alfred Lord Tennyson (1842)

Context & Meaning: Written after the sudden death of Tennyson’s close friend, this dramatic monologue explores aging, purpose, and the refusal to let old age or the approach of death diminish a person's inner drive.

Key Plot & Mechanics: An aging King Ulysses (Odysseus) sits safely at home but feels completely restless and bored by his quiet life. He gathers his old crew of sailors for one final, dangerous voyage into the sunset. He acknowledges that they are old and that death closes everything, but he insists on living fully until that moment arrives, famously declaring his will *“to strive, to seek, to find, and not to yield.”*

Theme: The poem highlights the human struggle against stagnation in the final chapters of life, showing that true vitality comes from pursuing adventure and purpose until the very end.

“Because I could not stop for Death” by Emily Dickinson (1890)

Context & Meaning: This calm, brilliant poem re-imagines the terrifying figure of the Grim Reaper as a courteous, unhurried gentleman taking the speaker on a carriage ride out of life.

Key Plot & Mechanics: Because the speaker is too busy with daily life to stop for death, Death kindly stops his carriage to pick her up. They take a slow, peaceful drive past a schoolyard of

playing children, fields of grain, and the setting sun—representing the stages of life. They finally stop at her new "house" in the ground (a grave), and the speaker reveals that centuries have passed since that ride, yet it feels shorter than a single day.

Theme: The poem highlights the peaceful, inevitable transition from mortal life to eternity, stripping away the traditional horror of dying by presenting it as a quiet, dignified journey.

“Gitanjali 95” by Rabindranath Tagore (1913)

Context & Meaning: This deeply spiritual poem uses the transition of birth to explain and comfort the human anxiety surrounding death, framing the unknown future with trust.

Key Plot & Mechanics: The speaker reflects on the profound moment they first entered the world at birth, realizing that despite being a total stranger to life, they were immediately met with love, light, and protection. Using this logic, the speaker concludes that the next great unknown—death—must be exactly the same. Because they loved this life, they trust that they will love whatever form comes next.

Theme: The poem highlights death as a continuation of a beautiful cycle, showing that the same cosmic grace that welcomed us into birth will comfort us when we depart.

15th Symphony: Fourth Movement by Dmitri Shostakovich (1972)

Context & Meaning: Written while the composer was facing severe illness and his own impending mortality, this complex orchestral piece serves as a haunting, personal meditation on life, death, and musical legacy.

Key Plot & Mechanics: The movement opens with a somber, ominous brass theme that references Wagner's "Fate" motif. Shostakovich weaves in musical quotes from his own past works alongside a ticking percussion rhythm that sounds like a mechanical clock or a fading heartbeat. The piece doesn't end with a grand, dramatic explosion, but rather fades away into a quiet, eerie stillness of clicking percussion instruments.

Theme: The piece highlights the stark reality of time running out, capturing the quiet, uneasy acceptance of an artist looking back at their life's work as the clock ticks down.

“Please Call Me by My True Names” by Thich Nhat Hanh (1976)

Context & Meaning: Written by a Zen Buddhist monk during the Vietnam War era, this profound poem explores the concept of interbeing—the idea that all living things, good and bad, are deeply interconnected.

Key Plot & Mechanics: The speaker describes themselves existing simultaneously as multiple conflicting figures: a transformation bud on a spring branch and a bird with crippled wings; a wealthy child and a starving kid in Uganda; a ruthless sea pirate raping a refugee girl and the

young girl herself who jumps into the ocean. By embracing every joy and every tragedy as part of their own identity, the speaker wakes up to true compassion.

Theme: The poem highlights the illusion of separation between human beings, showing that true spiritual maturity requires recognizing ourselves in both the suffering and the cruelty of the world.

“A Letter from Isaac Asimov to his Wife Janet...” by David Berman (1999)

Context & Meaning: This poem is a moving, tender look at the collision between cold, scientific rationalism and the deeply emotional reality of saying goodbye to someone you love.

Key Plot & Mechanics: Written from the perspective of the famous sci-fi writer on his deathbed, the speaker addresses his wife. As a lifelong secular rationalist, he doesn't offer comfort through religious myths or an afterlife. Instead, he grounds his love in the physical universe, looking at his wife's face as the only truth that matters while his own conscious mind prepares to dissolve back into the elements of nature.

Theme: The poem highlights how love provides profound meaning even in a strictly material universe, showing that acceptance of death doesn't require fantasy, just deep gratitude for the connection shared.

“In the End” by Linkin Park (2000)

Context & Meaning: This iconic nu-metal rock song captures the intense frustration, exhaustion, and ultimate resignation that comes with pouring all your effort into a relationship or a goal, only for it to fail.

Key Plot & Mechanics: The song alternates between rapid, anxious verses tracking the relentless, unstoppable ticking of time and a heavy, melodic chorus. The speaker describes trying incredibly hard, pushing through the pain, and sacrificing everything to make things work. However, the crushing reality hits in the famous refrain: *"I tried so hard and got so far, but in the end, it doesn't even matter."*

Theme: The song highlights the painful futility that humans often feel when facing failure, showing the existential realization that despite our best efforts, time and circumstance can wipe away everything we build.

Home and wandering

Literature & Poetry on Immigration and Identity

- Richard Blanco's **"Mother Country"** (2019) explores the dual identity of a Cuban immigrant through her child's eyes, detailing the emotional weight of leaving her homeland in 1968 and the resilience required to build a new life in America.
- The poem captures the sensory blend of culture—symbolized by pairing pernil with mac and cheese—and highlights the sacrifice of loving a country by choice, concluding that one's true home is defined by where a person chooses to claim their final place.
- Adrian Castro's **"The Sound of One Immigrant Clapping"** reimagines the immigrant journey as a pursuit of musical and physical liberation in New York, contrasting Latin rhythms with the harsh reality of factory labor.
- The work depicts the struggle to preserve native melody and language amidst political trauma, illustrating the difficult process of letting go required for a newcomer to finally feel at home.
- Marilyn Chin's **"We Are Americans Now, We Live in the Tundra"** uses a blues song framework to mourn lost ancestral heritage alongside ecological decay.
- The speaker juxtaposes the sterile environment of American urban sprawl against vibrant memories of Chinese history, offering a poignant farewell to traditional scholars as an industrialized identity takes hold.
- Adrienne Rich's **"Immigrants Please Note"** presents a pivotal choice between remaining in a safe, known reality or stepping through a

door into a transformative state of heightened awareness.

- Choosing to stay allows for a dignified, stable life at the cost of spiritual blindness, while stepping through requires confronting one's true identity with no guarantee of simplicity.
- Emma Lazarus's "**The New Colossus**" contrasts the aggressive, conquering statues of the ancient world with the Statue of Liberty, redefined here as the compassionate "Mother of Exiles."
- Rather than boasting of military strength, this figure offers a beacon of welcome to those fleeing oppression, famously demanding that other nations send their most vulnerable to her shores.

Transit, Isolation, and Famous Terminal Case Studies

- The mobile application **Fog of the World** tracks physical exploration in real-time, covering unexplored areas on a digital map with clouds that clear away as the user travels.
- The film ***The Terminal*** depicts a traveler whose country collapses while he is in transit, invalidating his passport and leaving him stranded inside an airport terminal.
- The movie was inspired by **Mehran Karimi Nasseri**, an Iranian refugee who resided inside Paris's Charles de Gaulle Airport for 18 years after claiming his travel documents were stolen.
- Former NSA contractor **Edward Snowden** spent six weeks hiding inside Moscow's Sheremetyevo International Airport after leaking classified documents detailing global surveillance programs.

- Snowden eventually vanished from the terminal without a trace through a covert escape coordinated by the Russian government while journalists waited outside.

Historical Navigation Instruments & Methods

- The **compass** was invented in China—originally featuring "south-pointing spoons"—to align along Earth's magnetic north-south axis, though a gyrocompass relies on Earth's rotation to indicate true geographical north.
- The **astrolabe** served as an early star chart and sky-model precursor to the sextant, allowing navigators to measure the altitude of celestial bodies above the horizon.
- The **sextant** measures the angular distance between two visible objects for celestial navigation, a design principle found in the unpublished writings of Isaac Newton.
- Before electronic calculators, **logarithm tables** were used to simplify complex spherical trigonometry calculations by converting multiplication and division into addition and subtraction.
- **Ephemerides** are reference books containing calculated trajectory tables for astronomical objects and spacecraft, marking one of the earliest practical uses for modern computers.
- Ships historically determined their speed using a **log and line**, letting a wooden board pay out from a reel while counting the number of tied knots over a set time measured by an hourglass.

- The **marine chronometer** is an extremely precise mechanical timepiece invented specifically to solve the problem of calculating longitude at sea.
- Earth is mapped using a grid of **latitude lines** (running east-west from the equator) and **longitude lines** (running north-south and meeting at the poles), which intersect to create precise geographic coordinates.

Musical Interpretations of Migration

- Led Zeppelin's "**Immigrant Song**" (1970) frames migrants as Viking explorers crossing the ocean in search of gold and glory, treating the sea as a pathway to conquest rather than a place of sorrow.
- Genesis's "**Illegal Alien**" (1983) addresses the frustration of moving without official documentation, highlighting how a lack of legal status leaves individuals feeling invisible and trapped in limbo.
- Los Tigres del Norte's "**Somos Más Americanos**" (2013) challenges traditional definitions of migration, asserting that Latin American immigrants belong to lands historically inhabited by their ancestors regardless of modern borders.

Visual Art and Liminal Space

- Frida Kahlo's painting ***The Bus*** depicts various social classes sitting side-by-side in Mexican society while subtly referencing the catastrophic bus accident that altered her life.

- Kovi Konowiecki's photographic project ***Borderlands*** explores physical and internal boundaries, documenting individuals and communities existing in the space between belonging and abandonment.

Animal Navigation and Biological Systems

- **Magnetoreception** allows migratory birds and other organisms to sense Earth's magnetic field, often facilitated by blue-light-sensitive **cryptochrome** proteins in their eyes.
- **Olfactory navigation** is a behavioral model suggesting certain species, like passenger pigeons, construct mental maps based on the spatial distribution of environmental odors.
- Animals utilize **echolocation** by emitting sound waves—either at a constant frequency or through frequency modulation—and processing the returning echoes via the Doppler effect to navigate.
- **Zugunruhe** refers to the anxious, hyperactive behavior and altered sleep cycles displayed by migratory animals when held in captivity during their natural migration period.
- **Sunlight polarization** occurs as light passes through the atmosphere, restricting light waves to a single plane that many animals use as a directional compass.
- Spatial memory enables the brain to store and recall structural information about physical surroundings to navigate efficiently.
- **Indigo buntings** navigate night migrations from North America to Central America by learning the star patterns around the celestial

pole shortly after birth, as demonstrated in Stephen Emlen's planetarium experiments.

- **Dung beetles** in South Africa maintain a straight path while rolling dung balls by orienting themselves using the visible light of the Milky Way galaxy.
- **Honeybees** navigate back to their hives by following linear terrain landmarks, sensing light polarization, and utilizing olfactory cues.
- **Sea turtles** rely on distinct regional variations in Earth's magnetic field to navigate back to specific nesting beaches, developing more detailed internal maps as they age.
- **Desert ants** track their position across featureless terrain using a combination of internal step counters, sky compasses, and olfactory signals.

When were Going we will still need them

Historic Roads & Networks

- Under Emperor Trajan, the Roman Empire reached its greatest extent in A.D. 117 and created the first large integrated network of highways, most of which were unpaved and un-named but featured regular milestones.
- Constructed by Rome's first emperor, the Via Augusta linked the port city of Cádiz directly to Italy, while Rome also integrated existing desert camel paths after conquering Egypt.
- The Royal Road of Persia ran from Sardis near the Aegean Sea to Susa in modern-day Iran, operating long before Roman roads and allowing couriers to cover the vast distance in just nine days.
- The Inca road system connected Cusco to Colombia and Chile through rough mountain terrain, relying on running messengers called Chasquis because the Incas did not utilize wheeled vehicles.
- The Via Francigena served as a famous European pilgrimage route connecting Canterbury in England to Rome.
- The Trail of Tears refers to the forced march where the United States compelled Native Americans to leave their ancestral homelands and walk to modern-day Oklahoma.
- The Grand Trunk Road is a historic trade route stretching from Bangladesh to Afghanistan through India and Pakistan.
- The King's Highway ran from Egypt through Jordan into Syria, making it one of the oldest continuously used routes in the world.

Road Types & Construction Methods

- Pioneer John Loudon McAdam developed Macadam construction, which involves compacting layers of crushed stone and binding them with cement, bitumen, or tar.
- Standard pavement refers to the surface of a road covered with concrete or tarmac, while asphalt (or bitumen) is produced through the fractional distillation of crude oil.

- Corduroy or log roads were created by laying logs perpendicular to the direction of travel to cross swampy or muddy terrain.
- Gallery roads—also known as cliff or plank roads—were constructed in mountainous regions of China using wooden planks fastened directly into cliff sides.
- Ridgeways utilize natural hard surfaces or hilltops as narrow, low-maintenance, unpaved paths.
- Milestones are physical markers installed along routes to display the distance to a specific town or city, whereas trackways are paths worn into the ground over time by repeated foot traffic.

Future Road Technologies

- Future roads may incorporate piezoelectric materials that generate electricity from mechanical pressure, as well as inductive wireless charging that powers electric vehicles while driving.
- As of 2024, approximately 10 operational electric road systems used either in-ground or overhead power lines to supply energy to moving vehicles and trams.
- Dedicated Short Range Communications (DSRC) transfer data to and from vehicles using the 5.9 GHz band, primarily facilitating electronic toll collection.
- Self-healing asphalt, designed using Google Cloud AI and biomass waste, automatically repairs surface cracks to extend road life.
- Modular pavement consists of brick- or tile-like units that shorten construction time and simplify maintenance, while permeable pavement uses porous materials to allow water drainage and prevent pooling cracks.

Human Behavior, Desire Paths, & Navigation

- Desire paths form when people take shorter or more convenient routes instead of designated walkways, reflecting a natural human preference for choice over confinement.

- Michigan State University planned its paved walkways by observing established desire paths, and Broadway in New York represents an early Native American desire path (the Wickquasgeck path) that cuts diagonally across Manhattan's modern grid.
- Vector-based navigation explains why humans and animals often choose paths that point most directly toward a destination, as the brain uses less energy calculating straight vectors than calculating the absolute fastest route.
- The explore-exploit problem illustrates the decision-making trade-off between choosing a known outcome (exploit) or trying a new alternative (explore), a balance that typically shifts from exploration in childhood to exploitation in adulthood.
- Route elasticity measures how willingly commuters switch paths when conditions change, with high elasticity leading drivers to adapt quickly and low elasticity keeping them on familiar routes regardless of traffic.

Algorithmic & Mathematical Routing Concepts

- The Travelling Salesman Problem focuses on finding the shortest possible path between multiple cities, serving as a key benchmark for optimization algorithms despite slowing down exponentially as cities are added.
- Similar to the Travelling Salesman Problem, the Steiner Tree Problem seeks the shortest possible network required to connect a specific set of points.
- Braess's Paradox, discovered by Arthur Pigou and named after Dietrich Braess, demonstrates that adding new roads to a network can actually slow overall traffic because drivers rush to use the new route.
- Dead reckoning estimates a current position by calculating speed and elapsed time from a previously known location.

Poems about travel

Du Fu travelling again 761 CE

Context: Du Fu is one of China's greatest poets. Writing during the chaotic aftermath of the An Lushan rebellion he spent much of his later life displaced and wandering

Meaning: Reflects the weariness of constant movement. Unlike a modern vacation, travel here is born of necessity and exile; it captures a sense of restlessness, the beauty of the passing landscape and a deep longing for stability and peace.

William Shakespeare Sonnet 50 1609

Context: Part of Shakespeare's famous sequence of sonnets often addressed to a beloved friend or youth

Meaning: Focuses on the mystery of travelling away from someone you love. He describes how heavy and slow the journey feels. Even his horse seems to understand his grief groaning under the weight of the poet's sadness because every step takes him further from his happiness

John Masefield Sea fever 1902

Context: He was an English poet who actually went to sea at a young age and this is one of his most famous works

Meaning: The poem is pure wanderlust. It expresses an uncontrollable almost biological urge to return to the ocean. It treats travel as a source of ultimate freedom adventure and companionship among sailors

Carl Sandburg Window 1914

Context: Sandburg was an American poet known for capturing industrial everyday life.

Meaning: Window is a brief vivid snapshot taken from a moving train at night. It contrasts the dark passing landscape outside with the brief glimmers of human life seen through windows. It captures the modern fleeting nature of travel where you are a momentary observer of other people's lives.

Robert Frost The road not taken 1915

Context: One of the most famous American poems

Meaning: Here the journey is an extended metaphor for life choices. Standing at a fork in the woods, the traveler must choose one path over the other. It is less about a road being inherently better and more about how the choices we make shape our entire life story knowing we can rarely come back and try the other path.

William Butler Yeats Sailing to Byzantium 1927

Context: Written when the Irish poet was in his sixties feeling the heavy onset of old age

Meaning. This is a spiritual and metaphorical journey. Yeats leaves behind the physical world of youth and decay (That is no country for old men) and sails to Byzantium a symbol of eternal art intellect and the soul. It is a journey away from the biological body toward timeless immortality

Rustu Onur The journey early 20th century

Context: He was a brilliant Turkish poet who died tragically young 22 from tuberculosis

Meaning: Given his short life his takes on journeys often carry an underlying weight of mortality transition and longing. The poem typically touches on the bittersweet nature of leaving things behind and the emotional baggage we carry when we move from one state of being to another

Czeslaw Milosz This Only 1988

Context: He was a polish poet who survived ww2 lived through communist rule and spent decades in exile in the US

Meaning: In his later poetry journeys become a way to look back at a long life. This only deals with memory and the realization that at the end of all our physical travels through time and space what remains is a desire to capture the absolute essence of past moments and the people we lost along the way

Richard Russo Just drive 1992

Context: Russo is an American novelist famous for writing about blue collar towns and family dynamics.

Meaning: This piece captures the psychological relief of the open road. Just driving is an act of escape a way to outrun problems clear the mind and find solace in the simple rhythmic act of motion without a strict destination

Rita Dove Vacation 1994

Context: He is a former US poet known for finding deep meaning in ordinary moments.

Meaning: this poem looks at the concept of modern travel through a sharp observant lens. It often focuses on the weird spaces of travel like airport waiting rooms where people drop their usual guards. Highlights how vacations promise relaxation but often just make us hyper aware of our own internal thoughts and the strangers around us.

Music about being on the road

Nat King Cole Get Your Kicks on Route 66

The song is like a musical map. Lists famous cities and tells the listener that the best way to see the world is by driving. Focuses on the adventure and the cool factor about car culture.

It makes a long dusty drive across the desert sound like a non-stop party. Ignores the boring parts of driving like being tired or traffic.

Willie Nelson On the road again 1980

This song is about the life of a touring musician. For him the road isn't just a vacation it is his home it reflects friendship and the love of seeing new things every day.

He calls his friends a band of gypsies and says he can't wait to get back on the road. It's a very happy view of a lifestyle that can actually be very exhausting and lonely for some people.

Tom Cochrane Life is a Highway 1991

In this song road isn't just asphalt it represents life itself. It reflects determination and energy. The idea is that life has ups, downs and long stretches but you have to keep driving.

It's a loud energetic saying that as long as you keep moving everything will be okay. It is a pump up song rather than a realistic look at a road trip.

Bic Runga Drive 1996

It's a much slower and thoughtful song. It's about two people in a car but it feels a bit lonely and uncertain. Reflects the quiet moments inside a car where you have time to think about your feelings.

It doesn't pretend that driving is always exciting but it shows that being on the road can be a time for deep thinking or even sadness

Norah Jones Come away with me 2002

This song treats travel as a way to hide from the rest of the busy world. Reflects the desire for peace and privacy. It's not about the car it's about where the car can take you.

Idealizes the escape. It's a very soft pretty song that makes travelling seem like a way to solve all your problems by simply leaving them behind.

Call Of Duty Free

Tourism and Trade

Dark Tourism

Dark tourism involves visiting places associated with death, tragedy, or other disturbing historical events. Examples include the Paris Catacombs and Alcatraz Prison.

Wellness Retreats

Wellness retreats focus on improving a person's physical and mental health. Activities often include spa treatments, meditation, yoga, and detox diets.

Edutourism

Edutourism, also known as educational tourism, involves travelling to learn about different cultures, histories, or subjects. The Grand Tour is one of the earliest examples of edutourism.

Voluntourism

Voluntourism combines tourism with volunteering, allowing travellers to contribute to local communities or conservation projects while visiting a destination.

Monastic Retreats

Monastic retreats are a niche form of tourism in which visitors stay in a monastery and experience the daily lives and routines of monks.

Agritourism

Agritourism involves visiting farms or agricultural attractions to learn about farming and food production. Examples include vineyards, tea plantations, and orchards.

Gastronomy Tourism

Gastronomy tourism, also known as culinary tourism, involves travelling to experience local food and cuisine. The concept became widely known after Erik Wolf published a white paper on the subject.

Ecotourism

Ecotourism is nature-based tourism that aims to conserve the environment while minimizing negative impacts. However, the term is sometimes used simply as a marketing buzzword for nature tourism that provides little environmental benefit.

Heritage Tourism

Heritage tourism involves visiting places of historical, cultural, or architectural significance to learn about their history and traditions.

Accommodation

Caravanserais

Caravanserais were fortified inns built along the Silk Road to provide accommodation, food, and protection for merchants and travellers. They were especially common across Asia and North Africa.

Hotels

The word “hotel” comes from the French word meaning “mansion.” Hotels became popular during the 17th century and offered more specialised services than traditional inns. The Grand Hotel in London is considered one of the earliest examples of a modern hotel.

Nishiyama Onsen Keiunkan

Nishiyama Onsen Keiunkan, located in Japan, is the world’s oldest hotel and the oldest continuously operating business. Although it has been renovated and rebuilt several times, it has remained in operation for centuries.

The Grand Tour

The Grand Tour was an expensive educational journey taken mainly by young English aristocrats. The traditional route passed through France, Italy, Germany, the Netherlands, Belgium, and then returned to England. It is considered a classic example of edutourism. The French Revolution and the Napoleonic invasions of Italy eventually disrupted the route.

Mass Tourism

Mass tourism developed as transportation became faster, cheaper, and more accessible. Although it made travel available to many more people, it can also disrupt local communities and cultures.

“The Golden Age of Travel Starts With You”

“The Golden Age of Travel Starts With You” is a campaign launched by the U.S. Department of Transportation to encourage courtesy and respect during air travel. It was introduced in response to increasing incidents of passenger misbehaviour and encourages travellers to dress appropriately, treat others respectfully, and appreciate the culture, traditions, and heritage of the places they visit.

Trade and Transportation

Entrepôt

An entrepôt is a trading centre where goods are imported, stored, and later exported or distributed. The word literally means “that which is placed between.” Entrepôts are usually located in strategic locations to reduce shipping time.

Way Station

A way station is a stopping point or transit location along a journey. In trade, it often serves as a transfer point for goods moving between destinations.

Free Trade Zone

A free trade zone is an area where goods can be imported, stored, manufactured, and exported without paying customs duties or taxes.

Transshipment

Transshipment is the process of transferring goods through an intermediate transit point before reaching their final destination. It is often used to combine several smaller shipments into one larger shipment, reducing shipping costs. For example, if there is little direct trade between Country A and Country C, goods may first be sent to Country B, where shipments are consolidated before continuing to Country C.

The Silk Road

The Silk Road was a vast network of trade routes connecting Asia with Europe. It was named after the valuable silk exported from China, although many other goods were traded along these routes. The Great Wall of China was built partly to protect important trade routes from invasion. Despite its name, the Silk Road was not a single road, and some historians argue that the name overlooks the contributions of many other trading civilizations.

Important Port Cities

Piraeus

Piraeus is the port city of Athens, Greece, and is the country's largest and busiest port. In ancient times, it served as Athens' main naval base. The Long Walls connected and protected the route between Athens and Piraeus.

Tyre

Tyre is an ancient port city in present-day Lebanon. It was originally an island but became connected to the mainland after Alexander the Great built a massive causeway during his siege of the city. Over time, the causeway widened because of silt deposits. Tyre was one of the most important Phoenician seaports.

Lothal

Lothal, located in present-day India, was part of the Indus Valley Civilization. It is believed to contain the world's earliest known dock and was an important trading centre. The name "Lothal" translates to "Mound of the Dead."

Carthage

Carthage, located in modern-day Tunisia, was the capital of the Carthaginian Empire and an important port city that developed from a Phoenician colony. According to legend, Queen Dido was granted as much land as an oxhide could cover. She cleverly cut the oxhide into thin strips and used them to enclose a much larger area, establishing the city.

Shimoda and Hakodate

Shimoda and Hakodate were the first Japanese ports opened to American ships under the Convention of Kanagawa in 1854. The treaty effectively ended Japan's long period of isolation. Before this, Japan conducted most of its foreign trade through Nagasaki with China and the Dutch.

Malacca

Malacca is a historic port city in Malaysia that was once one of the world's most important international trading centres because of its strategic location on the Strait of Malacca.

Modern Trade Hubs

Istanbul

Istanbul functions as a major entrepôt linking Asia and Europe. As trade routes shift, many goods travelling from China to Europe pass through Istanbul. The city is also one of the world's busiest aviation hubs.

Chicago

Chicago is one of the world's largest logistics and transportation hubs. It connects the Great Lakes with major rail, road, and air networks, making it an important way station for east-west and north-south trade.

Hong Kong

Hong Kong serves as an important gateway between China and the rest of the world. It operates as a free trade zone and has long been one of the world's busiest international ports.

Next Year in Futurism

Technologies That Are Always “Just a Few Years Away”

Fusion Power

Fusion power is considered an almost limitless and clean source of energy. It mimics the way the Sun generates heat by fusing two atomic nuclei together, releasing a large amount of energy in the process.

Fusion Energy Gain Factor

The fusion energy gain factor is the ratio of energy produced to the energy used. If the gain factor is equal to 1, the reactor produces exactly as much energy as it consumes. A value below 1 means more energy is used than produced. As of 2025, the National Ignition Facility in the United States is the only laboratory to have demonstrated a fusion energy gain factor greater than 1.

Cure for Cancer

Scientists have successfully cured certain types of cancer. However, cancer is not a single disease, and each type generally requires its own specific treatment or cure.

Graphene

Graphene is a form of carbon made up of a single layer of atoms arranged in a honeycomb pattern. It has extremely high tensile strength, excellent electrical conductivity, and is transparent. It is also the thinnest two-dimensional material ever discovered. When multiple layers of graphene are stacked together, they form graphite.

Nanotechnology

Nanotechnology involves materials and devices that measure between 1 and 100 nanometres. At this scale, they can interact directly with cells and DNA, making them useful in medicine and many other fields.

Space Elevator

A space elevator is a concept commonly found in science fiction. It consists of a cable stretching from Earth to geostationary orbit, where a counterweight is placed. The balance between Earth's gravity and centrifugal force keeps the cable under tension, allowing elevators to travel along it.

Food Pills

Food pills are a hypothetical technology designed to replace the need for regular meals by providing all the nutrients required for daily human nutrition.

Artificial General Intelligence (AGI)

Artificial General Intelligence, or AGI, refers to AI that can match human intelligence across many different tasks instead of being designed for only one. It is often described as having emotional and reasoning abilities similar to those of humans.

Extras

Take Me to the Year 2028

In 2018, several predictions were made about life in 2028. These included the replacement of mobile phones, half of all cars becoming autonomous, freelancers making up 75% of the workforce, nine generations being alive at the same time, the disappearance of cash, widespread use of solar power, athletes with artificial limbs becoming the fastest competitors at the 2028 Olympics, competitive gaming becoming an Olympic sport, and wireless charging becoming commonplace.

Wardenclyffe Tower

Wardenclyffe Tower was a project designed by Nikola Tesla as a prototype for wireless communication. Long tunnels were built beneath the tower, and Tesla hoped to use his Tesla coil to transmit information wirelessly over long distances.

Ways to Extend Life

Organ Transplants

Organ transplants can extend life by replacing damaged organs. Reports have claimed that Xi Jinping and Vladimir Putin discussed the possibility of reversing ageing through organ transplants. However, transplant recipients must usually take immunosuppressant drugs for life to prevent their immune system from rejecting the donated organ.

Rejection-Free Organs

Scientists are researching genetically edited pig organs created using CRISPR technology. These organs are designed to reduce the risk of rejection and have already been transplanted into humans in a small number of experimental cases.

Stem Cells

Stem cells are unspecialised cells that can develop into many different types of cells in the body. Researchers have even rebuilt a human thymus, an important part of the immune system, using stem cells.

Bryan Johnson

Bryan Johnson is an entrepreneur known for his efforts to reduce his biological age. One of his experiments involved receiving plasma from his 17-year-old son in the hope of slowing the ageing process.

Organoids

Organoids are small, simplified versions of organs that are grown artificially from tissue or stem cells. They are mainly used for medical research and drug testing.

Bioprinting

Bioprinting is similar to 3D printing but uses biological materials known as bio-inks to build tissues layer by layer. Although it can create tissue and organ models, it cannot yet produce fully functioning replacement organs.

Plasma Transfusion

Plasma transfusion involves transferring plasma from younger individuals, often under the age of 25, to older people in the hope of reversing ageing. However, there is currently no strong clinical evidence supporting these claims.

Senotherapeutics

Senotherapeutics are treatments designed to slow ageing by targeting cellular senescence, the natural ageing process of cells. This research is inspired by organisms such as the hydra, which show negligible biological ageing.

Cellular Reprogramming

Cellular reprogramming aims to reset the biological age of cells. One well-known method involves the Yamanaka factors. Scientists often use partial reprogramming to rejuvenate cells while allowing them to maintain their original function.

Telomere Extension

Telomeres are protective caps at the ends of chromosomes that shorten every time a cell divides. Extending telomeres could allow older cells to divide more times. However, telomere shortening also helps prevent damaged or mutated cells from reproducing, reducing the risk of cancer.

Caloric Restriction

Research suggests that reducing calorie intake without causing malnutrition may slow the ageing process by approximately 2–3%.

Digital Immortality

Digital immortality involves creating a digital version of a person, such as an AI clone or deepfake, so that aspects of their personality or appearance can continue after death.

Cryonics

Cryonics is the practice of freezing and storing a person's body after death in the hope that future medical technology may be able to revive and cure them. James Bedford was the first person to undergo cryonic preservation.

Extra

Retrofuturism

The term “retrofuturism” was first used in a book titled *Retro Futurism* by T. R. Hinchliffe. It explores how people in the past imagined the future and how modern society views those old predictions. The style became especially popular during the 1970s, when rapid technological developments such as computers and test-tube babies captured the public’s imagination. Retrofuturism combines vintage designs with futuristic ideas, and steampunk is one of its best-known genres.

Quantum Computing

Quantum Computing

Traditional computers store information as bits that can be either 0 or 1. Quantum computers use qubits, which can exist in multiple states at the same time because of quantum mechanics. These processors are extremely sensitive to environmental disturbances. Companies such as Google and IBM are competing to achieve quantum supremacy, which refers to solving problems that classical computers cannot solve efficiently. Google was the first company to announce that it had achieved quantum supremacy.

Sycamore

Sycamore is Google’s quantum computer, built around a 54-qubit processor. It completed a specific calculation in about 3 minutes and 20 seconds that Google estimated would take one of the world’s fastest classical supercomputers around 10,000 years. Google can also manufacture new quantum chips relatively quickly.

Cross-Entropy Benchmarking

Cross-entropy benchmarking is a method used to evaluate whether quantum computers are functioning correctly. It compares the results of quantum circuits with those generated by classical computers.

John Preskill

John Preskill is the physicist who coined the term “quantum supremacy.”

The Reality of Quantum Computing

Despite impressive demonstrations, current quantum computers remain far from being widely useful. They must operate at temperatures close to absolute zero (-273°C), and there are

currently very few practical applications that clearly outperform classical computers. However, sufficiently advanced quantum computers could one day break many existing encryption systems.

Q-Day

Q-Day is the hypothetical day when quantum computers become powerful enough to crack modern encryption methods, making many current password and security systems vulnerable.

AI Writing

Wikipedia's List of Signs of AI Writing

Wikipedia lists several characteristics commonly associated with AI-generated writing. These include avoiding phrases such as “it is far more complex,” overusing expressions like “serves as” or “stands as,” excessive use of bold text, frequent use of em dashes, and unusual reliance on tables.

Trojan Horse

A Trojan Horse technique involves hiding instructions inside a document to catch students who copy and paste the text into an AI tool. For example, an invisible instruction such as “write from the perspective of a 17th-century peasant” could be hidden in an essay about modern television.

Turing Test

The Turing Test was proposed by Alan Turing as a way to determine whether a machine can exhibit intelligence indistinguishable from that of a human. Passing the test is often associated with achieving AGI, although no AI system is universally accepted as having passed it.

Kola Superdeep Borehole

The Kola Superdeep Borehole is the deepest man-made hole ever drilled into Earth. An urban legend claims that scientists heard the screams of souls from hell coming from the borehole, but this story is fictional. The project began during the Cold War as part of a scientific race to drill as deeply as possible into Earth's crust. One objective was to obtain samples closer to the mantle. In places such as Oman, parts of the mantle are naturally much closer to the Earth's surface.

American Miscellaneous Society

The American Miscellaneous Society was an informal club made up of leading American scientists. It proposed the first serious attempt to drill to Earth's mantle through Project Mohole, which was named after the Mohorovičić discontinuity, the boundary between Earth's crust and mantle.

German Attempt at Digging

German scientists encountered difficulties drilling perfectly vertical boreholes. To solve this problem, they developed new drilling technologies that are now widely used in the oil and gas industry.